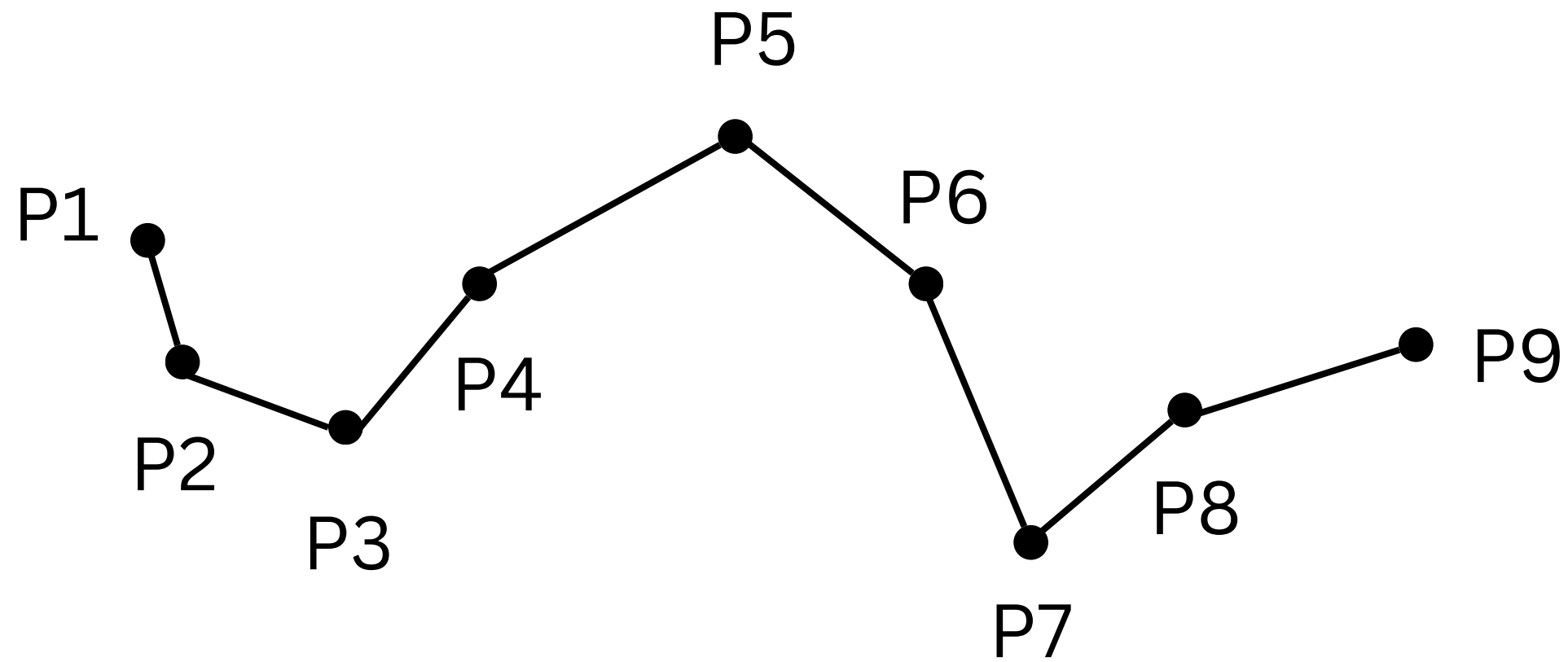


Ramer-Douglas-Peucker Polyline Simplification Algorithm

The Data Structures

 epsilon



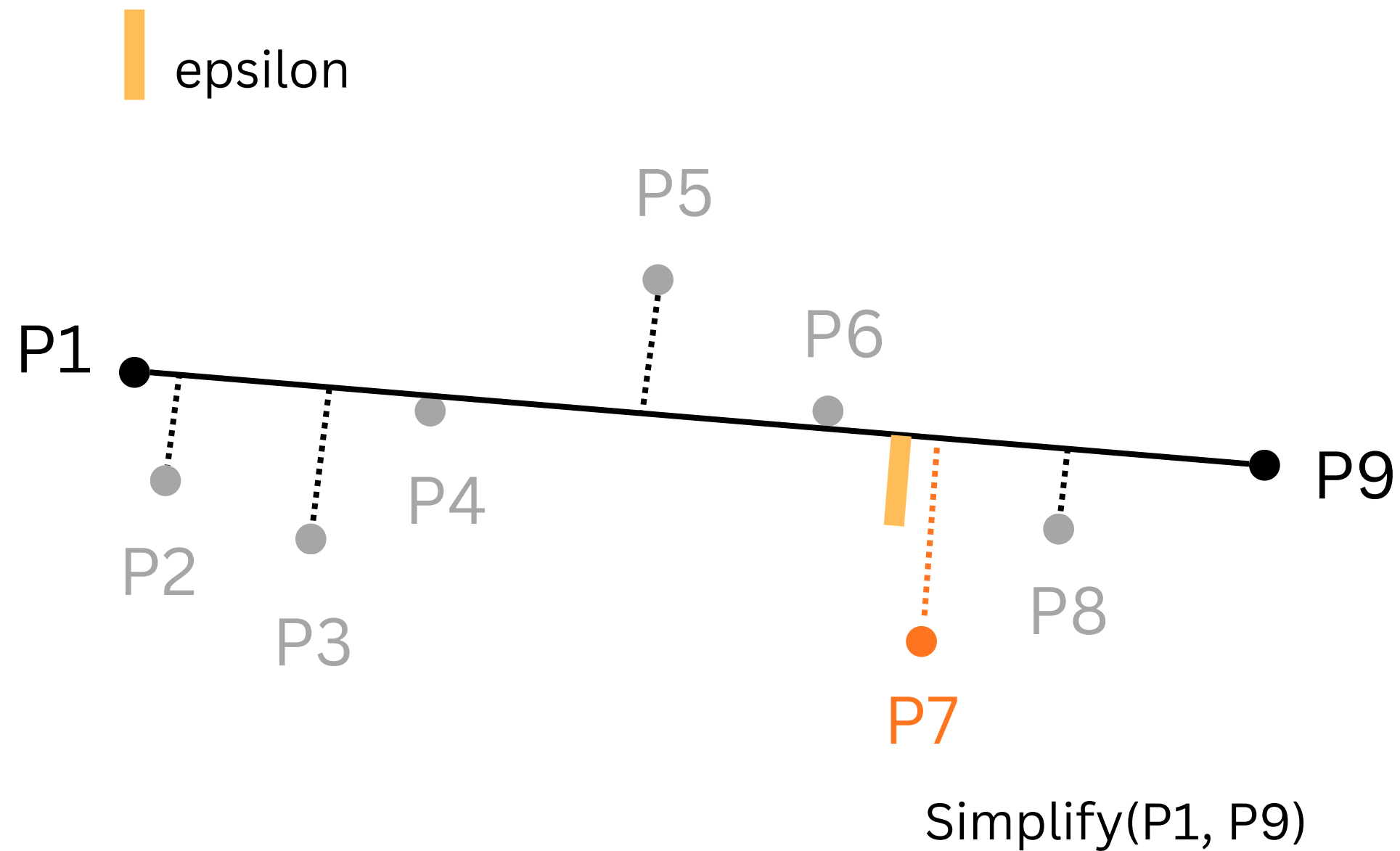
Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 P7 P8 **P9**

The Algorithm

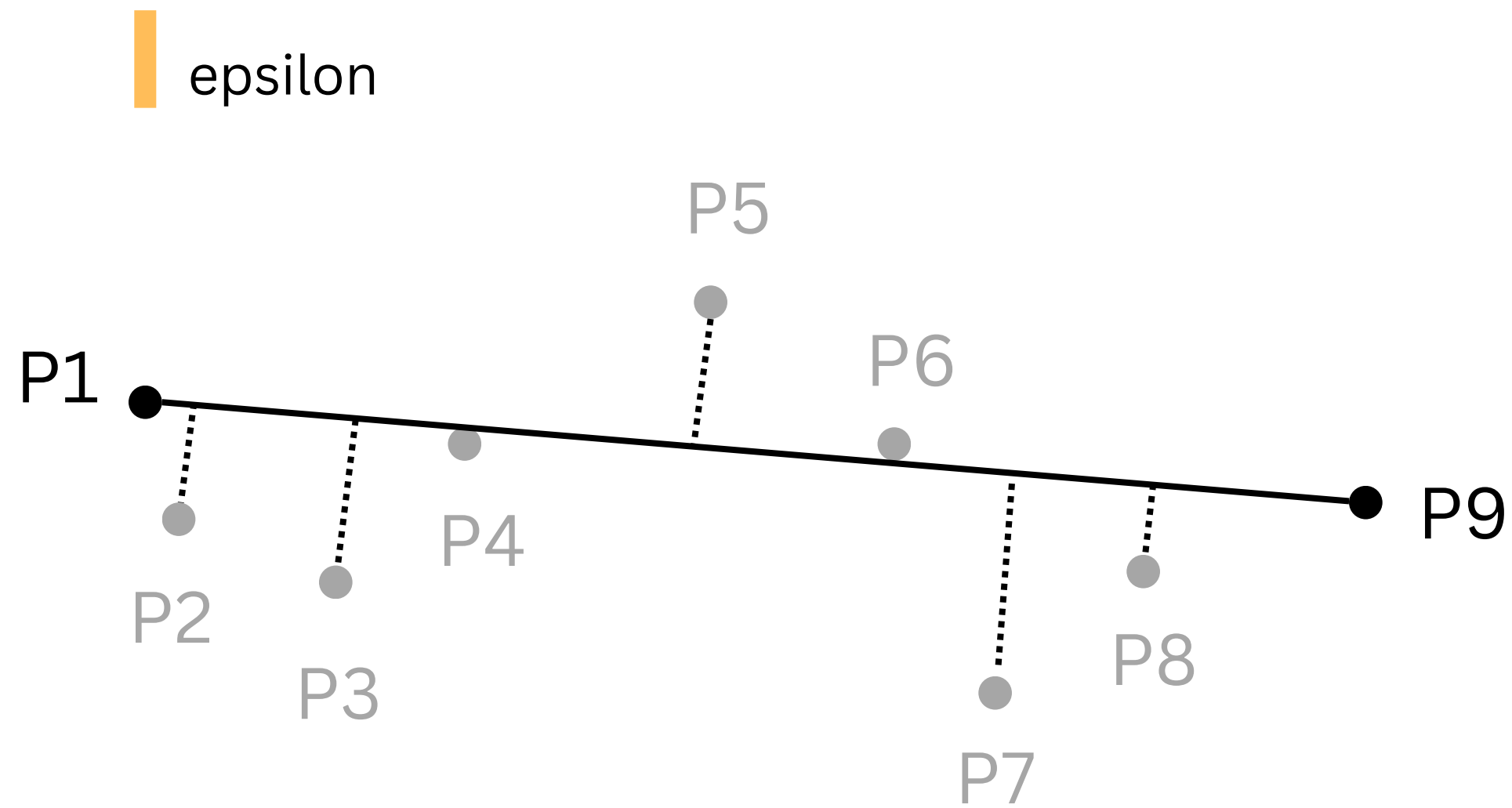


If the maximum of the perpendicular distances of the points not in the polyline from the current polyline is more than the epsilon length, you include the point, else you exclude it. On inclusion, you split the array of points into two halves and recursively check for the left and the right part.

Array

Call Stack

P1 P2 P3 P4 P5 P6 P7 P8 **P9**

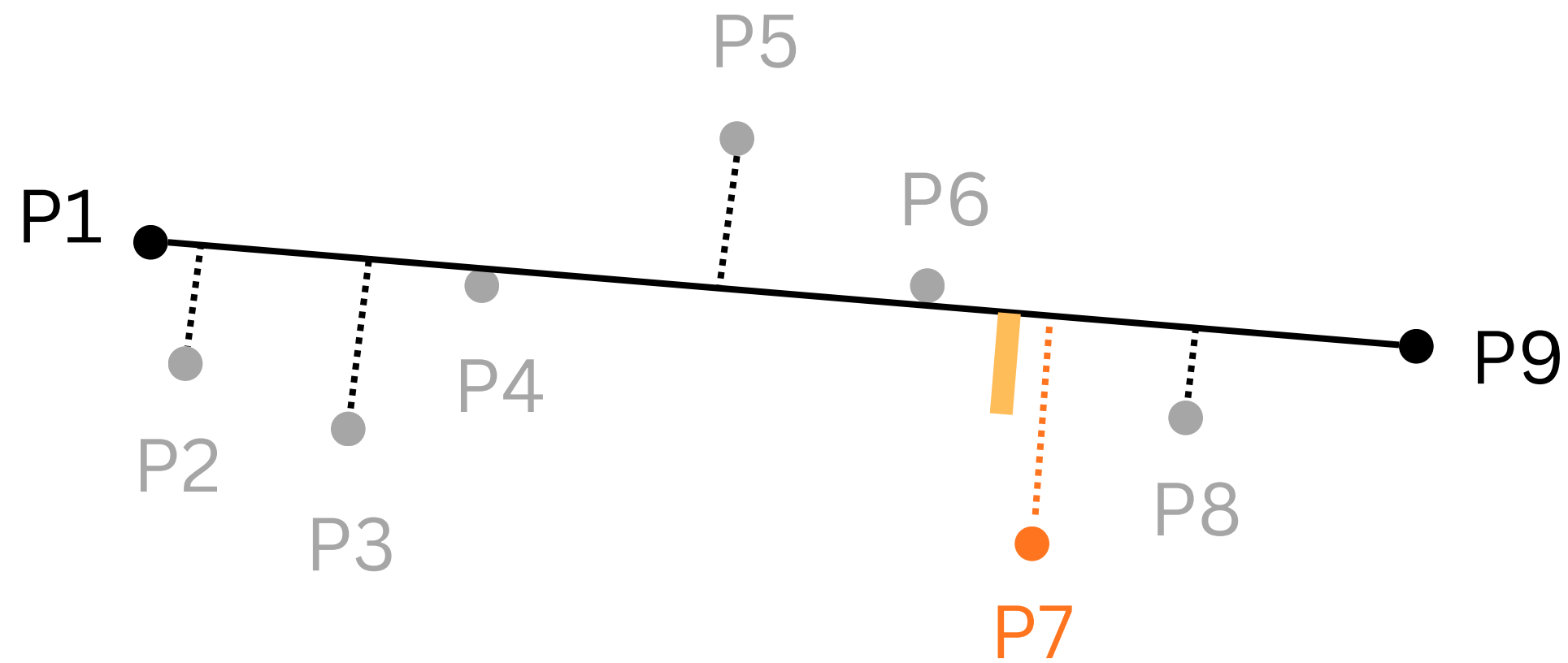


Simplify(P_1, P_9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 P7 P8 **P9**

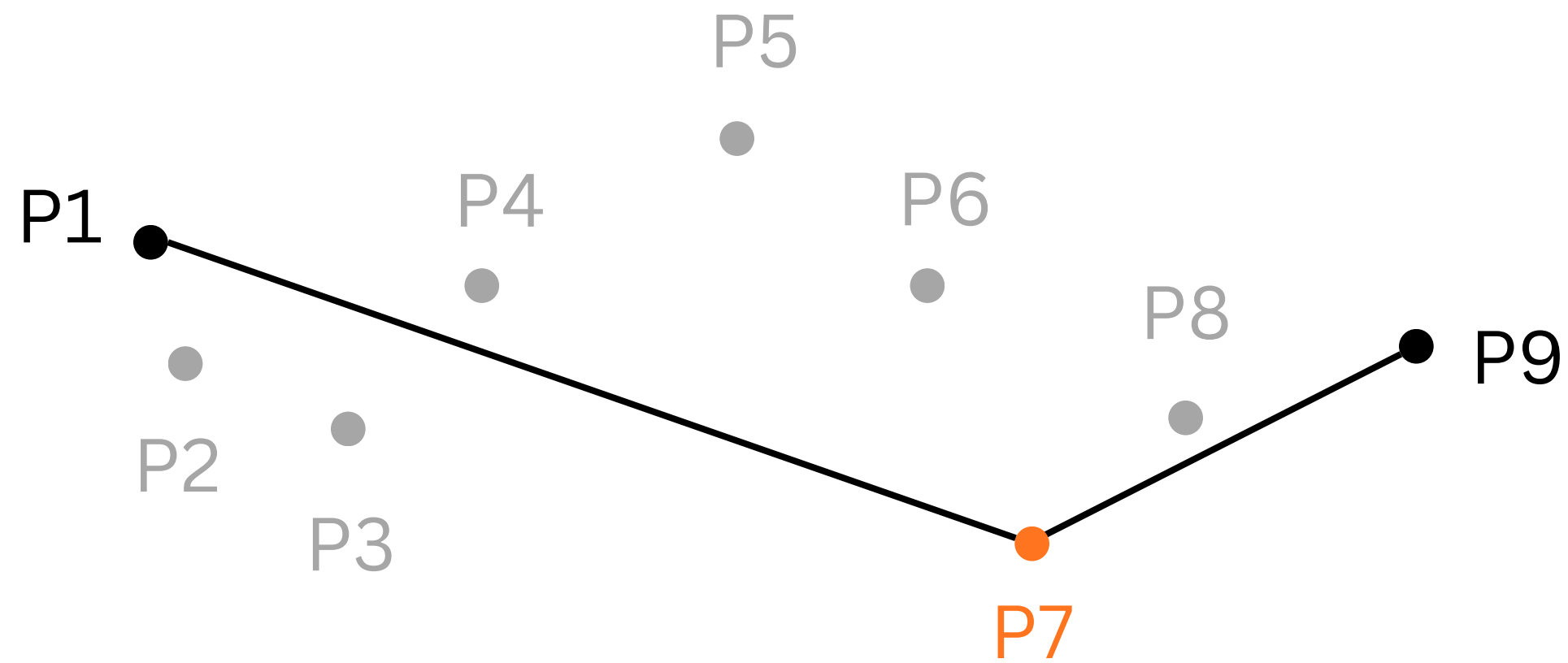


Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 **P7** P8 **P9**

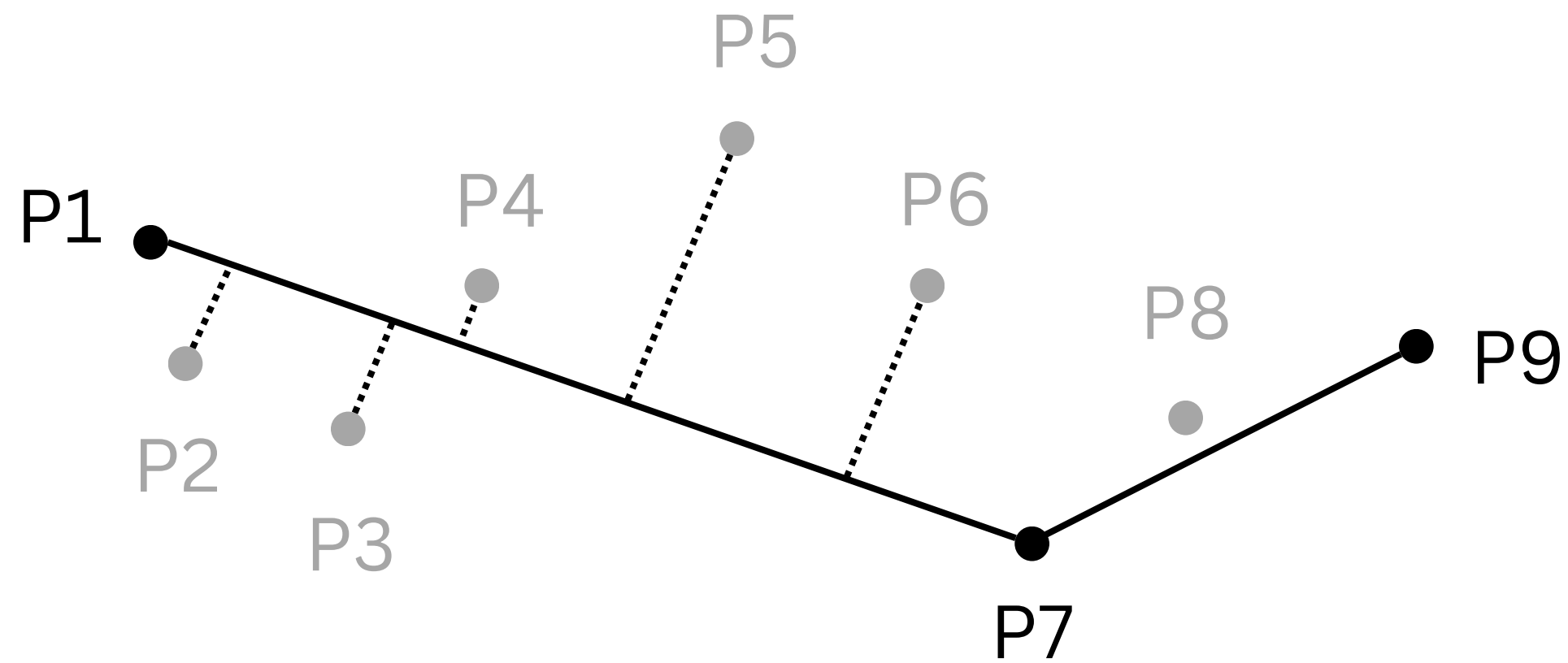


Simplify(P1, P7)
Simplify(P7, P9)
Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 **P7** P8 **P9**

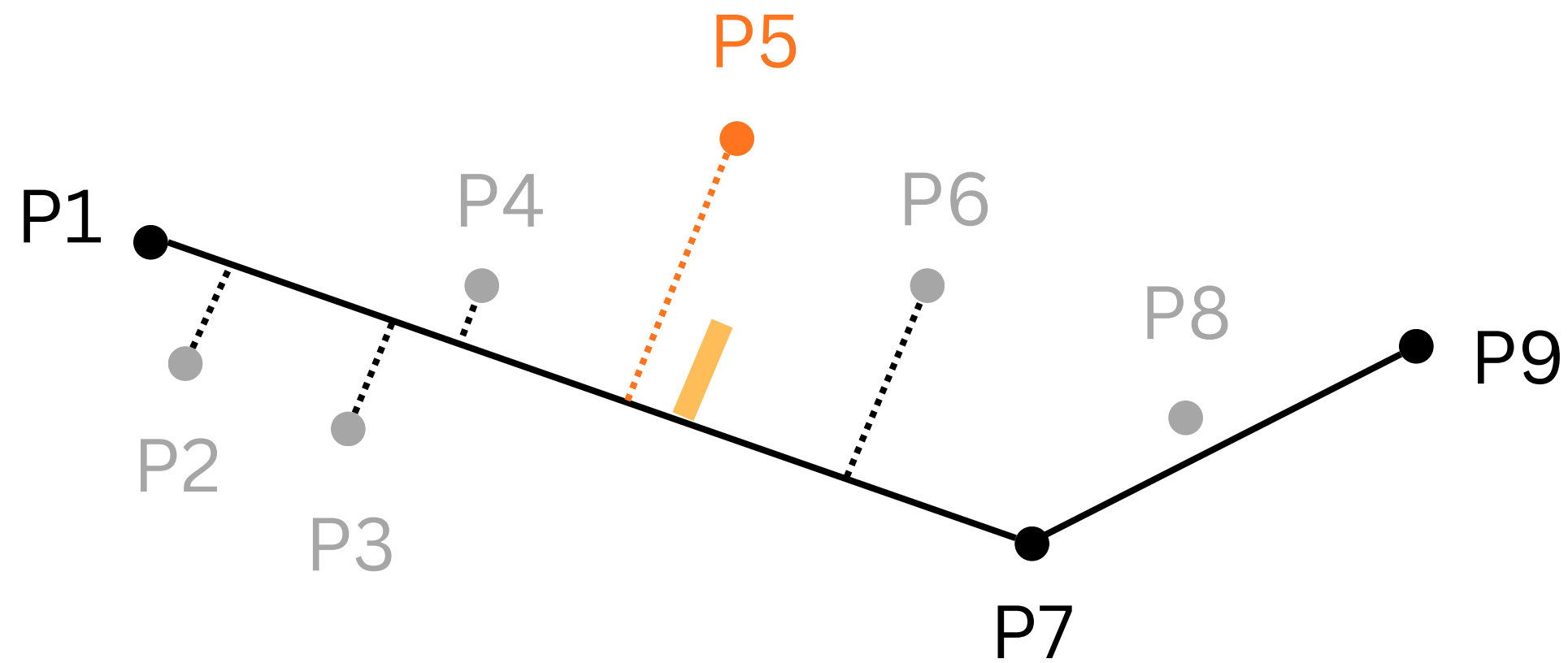


Simplify(P1, P7)
Simplify(P7, P9)
Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 **P7** P8 P9



Simplify(P1, P7)

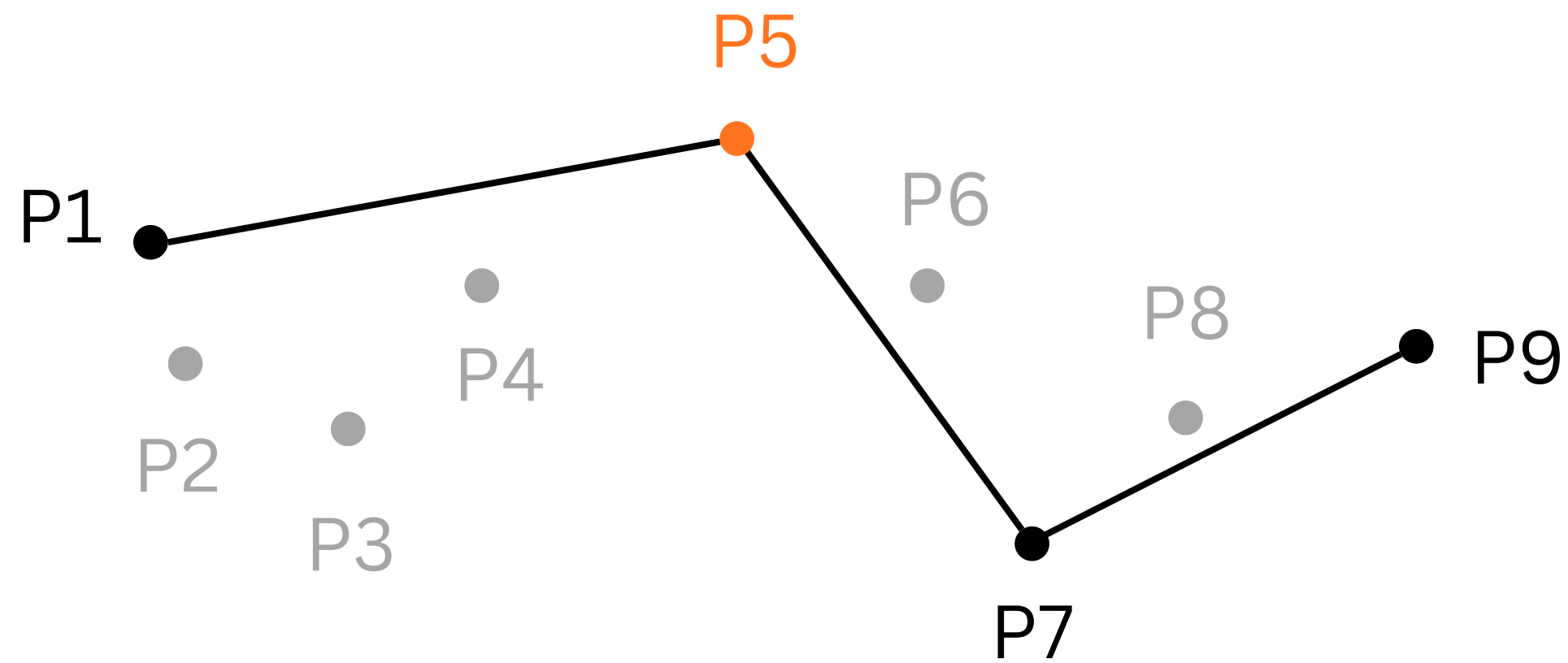
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 **P5** P6 **P7** P8 P9



Simplify(P1, P5)

Simplify(P5, P7)

Simplify(P1, P7)

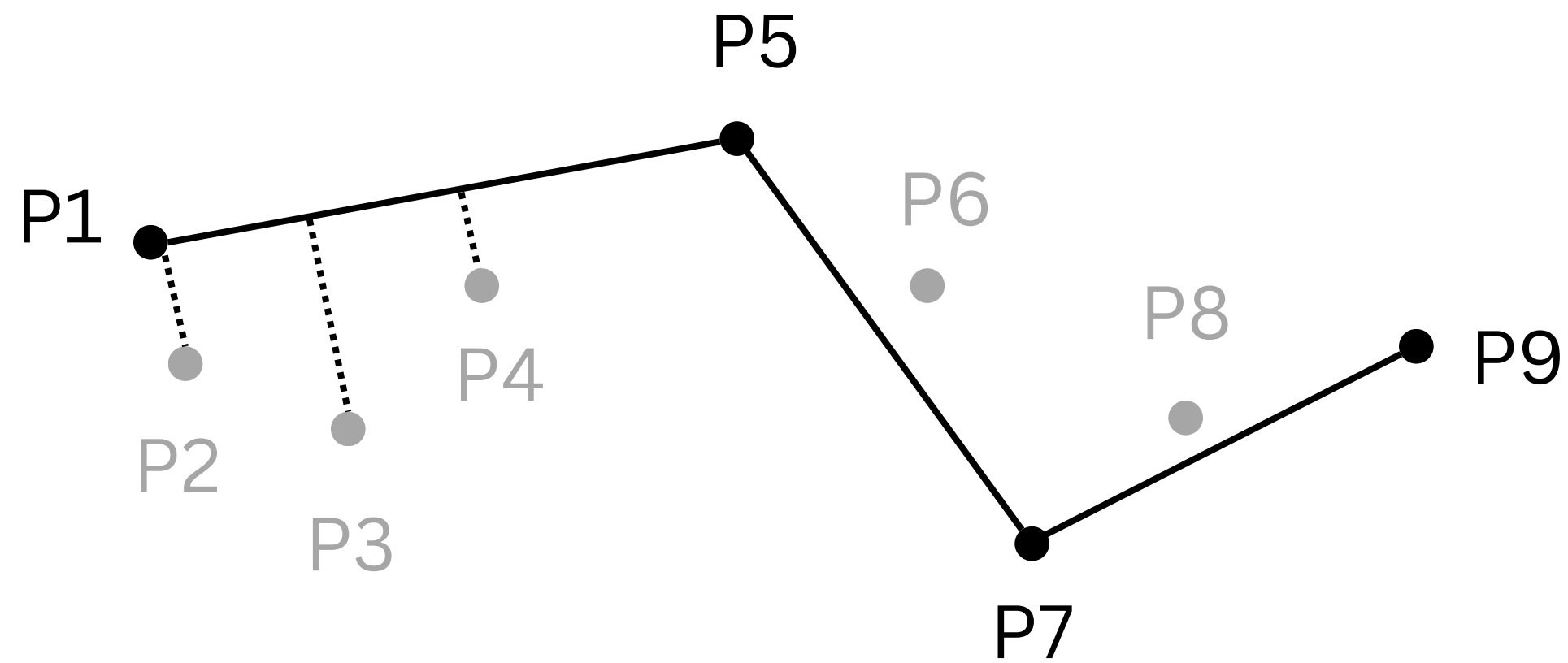
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 **P5** P6 **P7** P8 P9



Simplify(P1, P5)

Simplify(P5, P7)

Simplify(P1, P7)

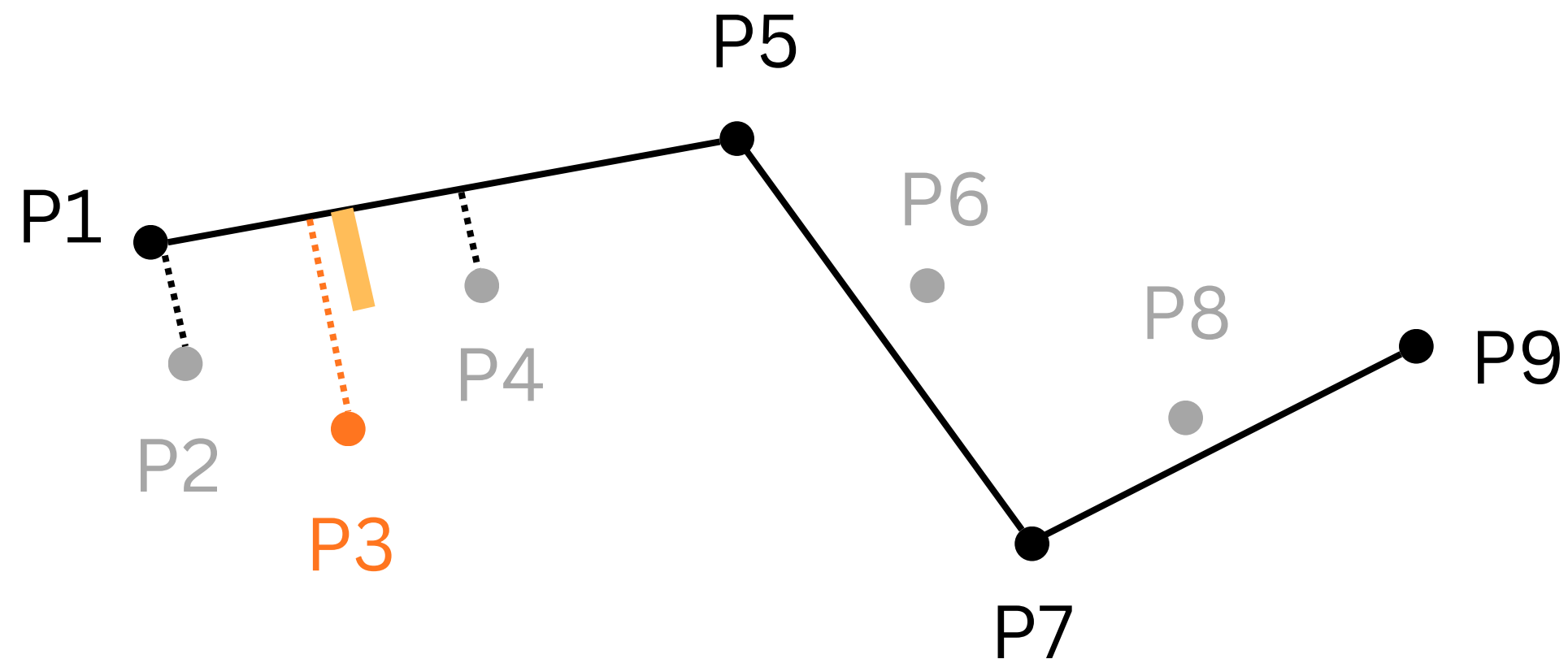
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 **P5** P6 P7 P8 P9



Simplify(P1, P5)

Simplify(P5, P7)

Simplify(P1, P7)

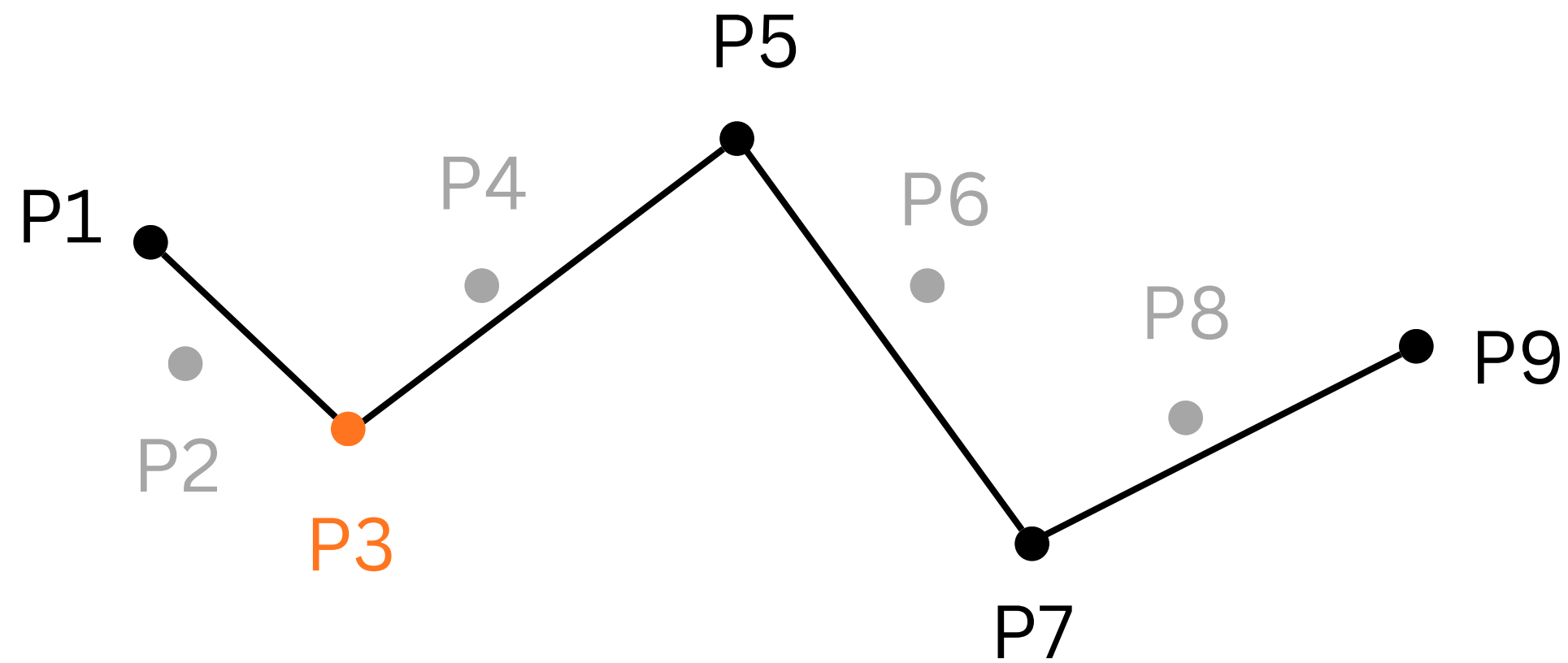
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 **P3** P4 **P5** P6 P7 P8 P9



Simplify(P1, P3)

Simplify(P3, P5)

Simplify(P1, P5)

Simplify(P5, P7)

Simplify(P1, P7)

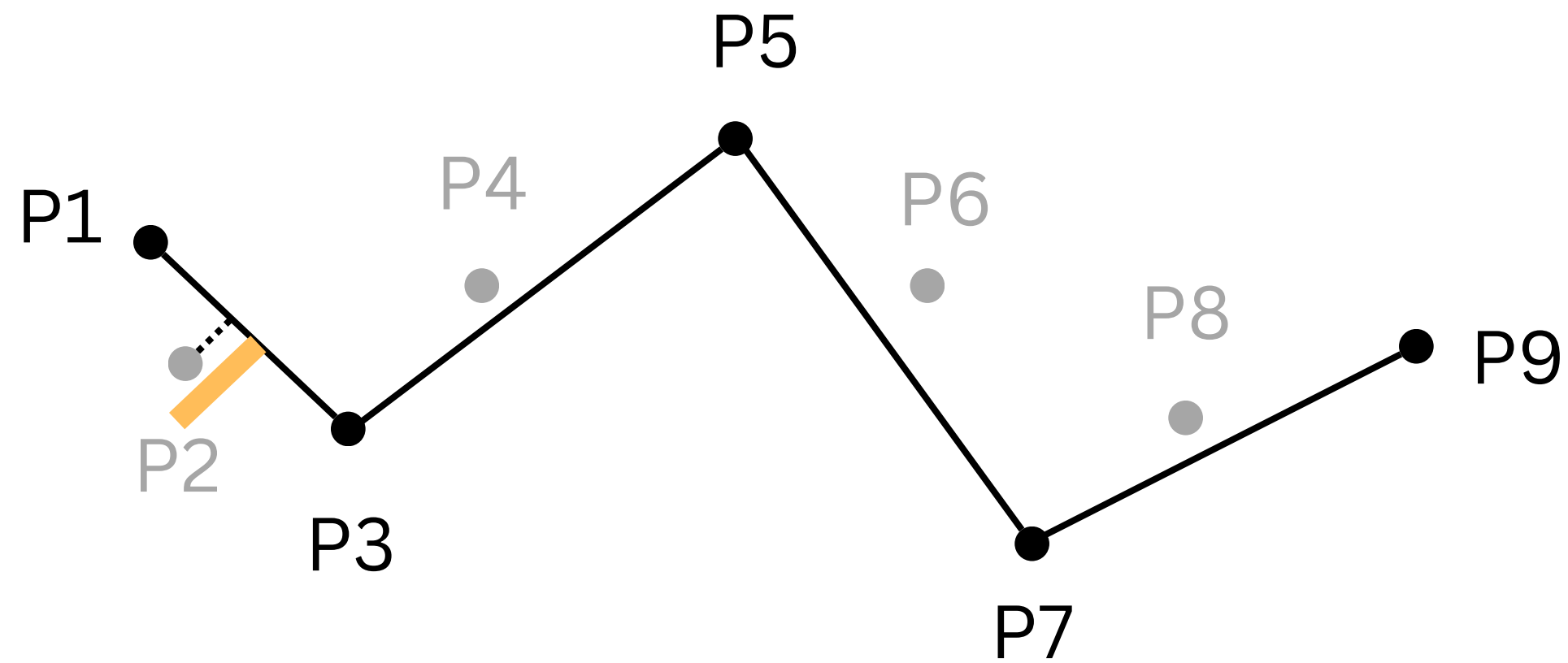
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 **P3** P4 **P5** P6 P7 P8 P9



Simplify(P1, P3)

Simplify(P3, P5)

Simplify(P1, P5)

Simplify(P5, P7)

Simplify(P1, P7)

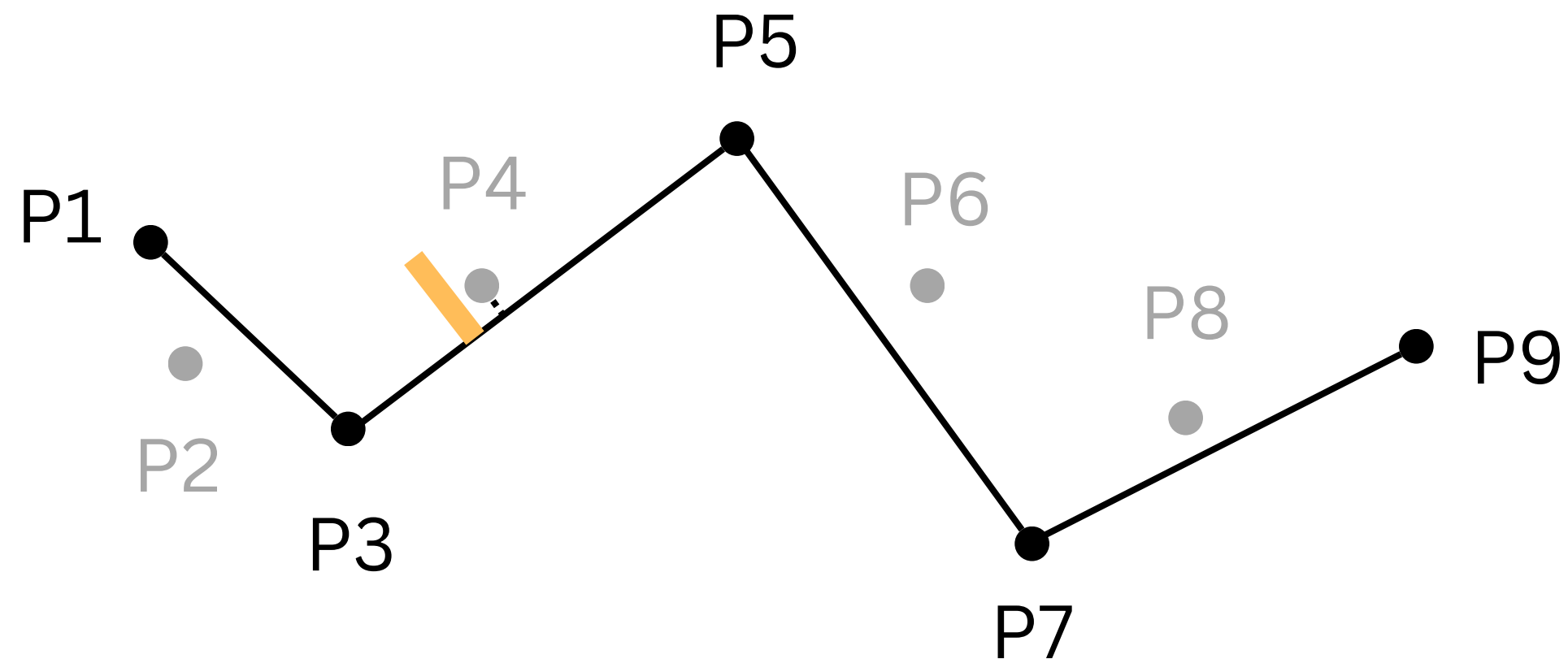
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 **P3** P4 P5 P6 P7 P8 P9

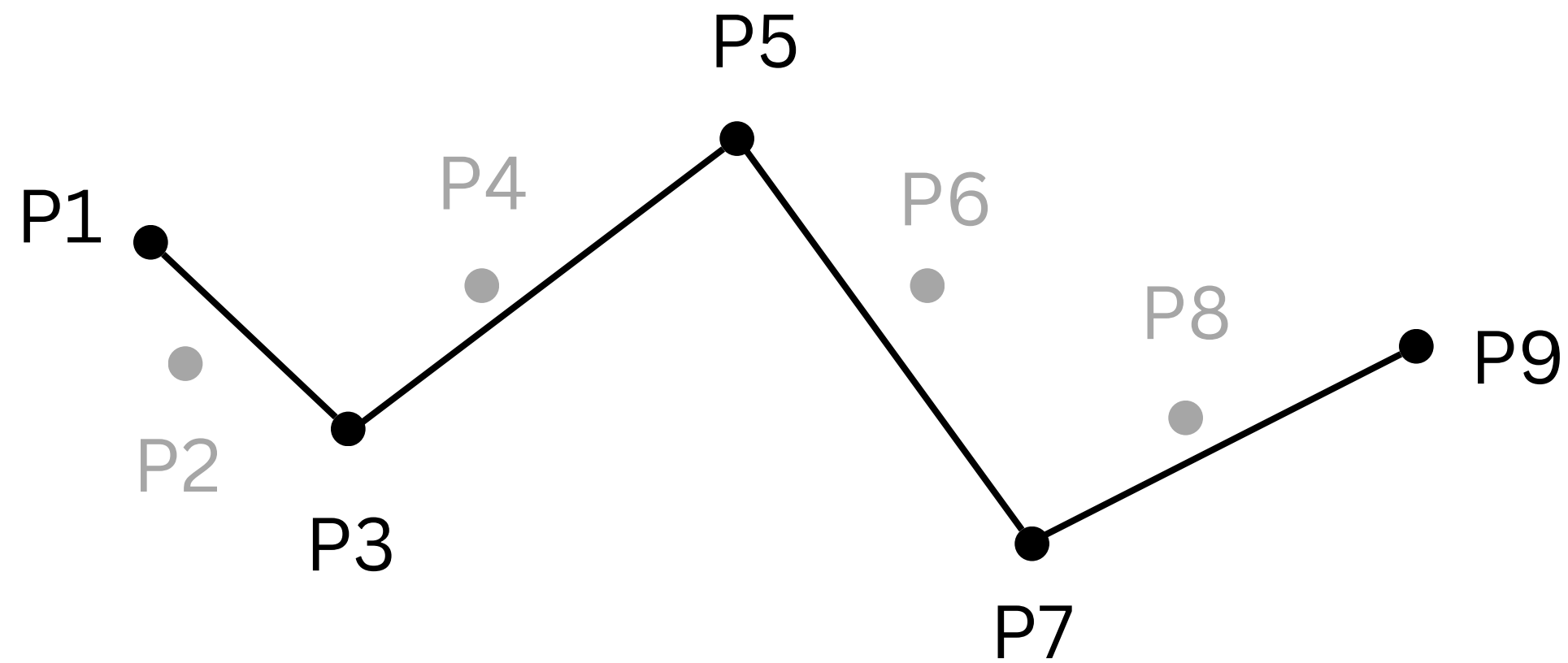


Simplify(P3, P5)
Simplify(P1, P5)
Simplify(P5, P7)
Simplify(P1, P7)
Simplify(P7, P9)
Simplify(P1, P9)

Call Stack

Array

P1 P2 **P3** P4 **P5** P6 P7 P8 P9



Simplify(P1, P5)

Simplify(P5, P7)

Simplify(P1, P7)

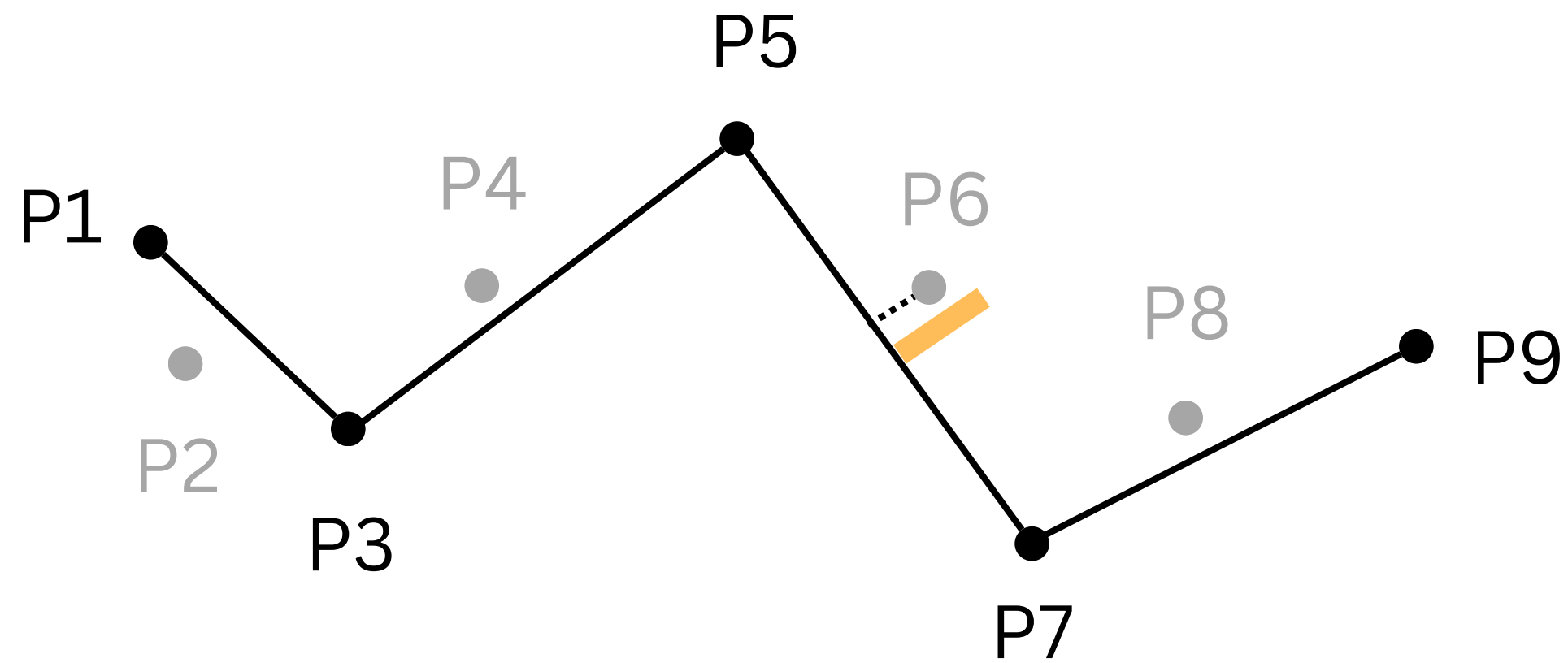
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 **P5** P6 P7 P8 P9

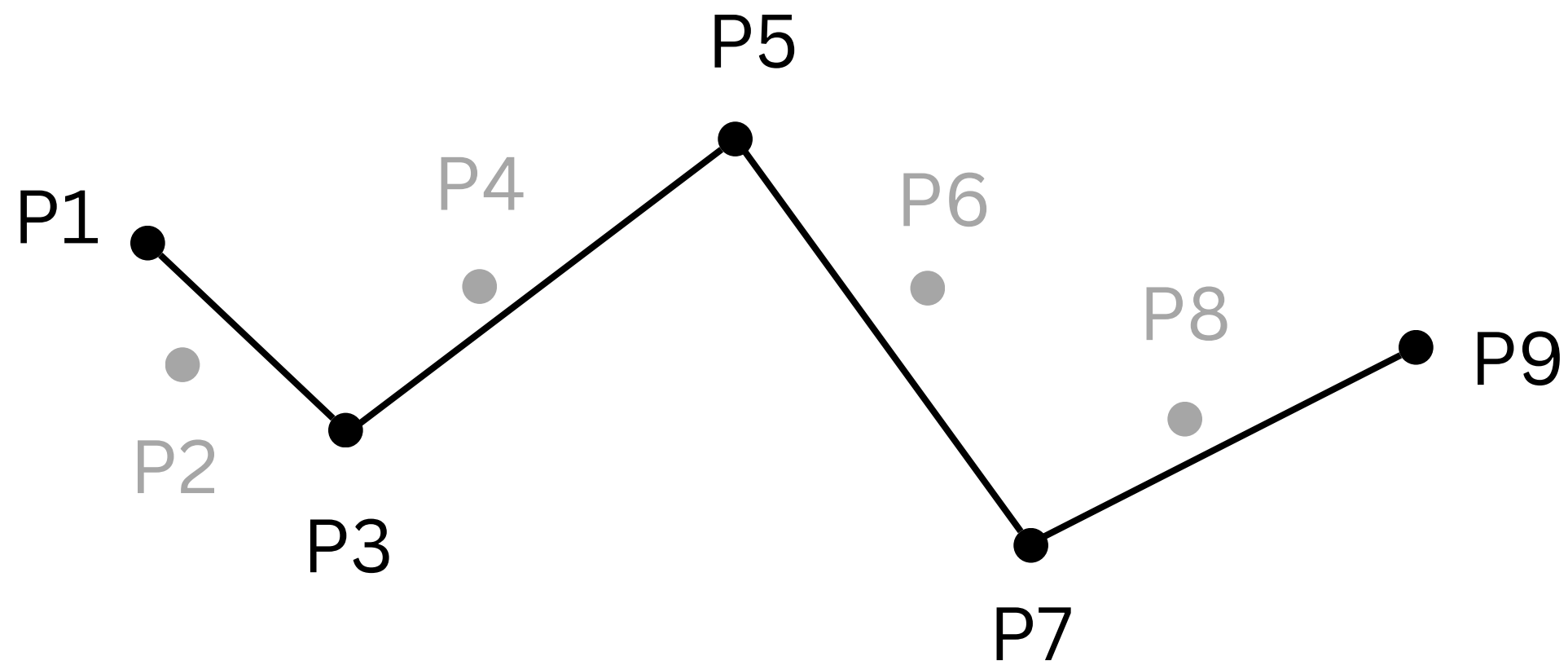


Simplify(P5, P7)
Simplify(P1, P7)
Simplify(P7, P9)
Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 **P5** P6 **P7** P8 P9



Simplify(P1, P7)

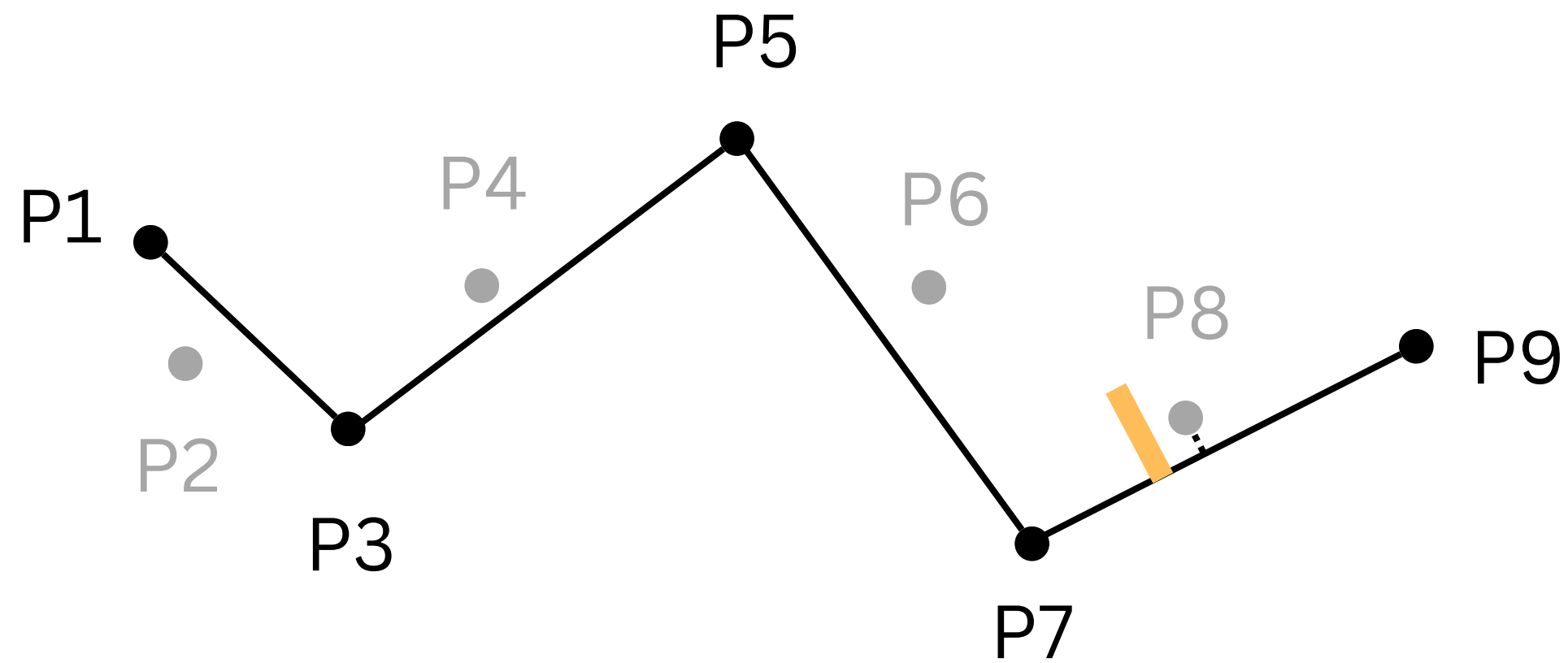
Simplify(P7, P9)

Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 **P7** P8 P9

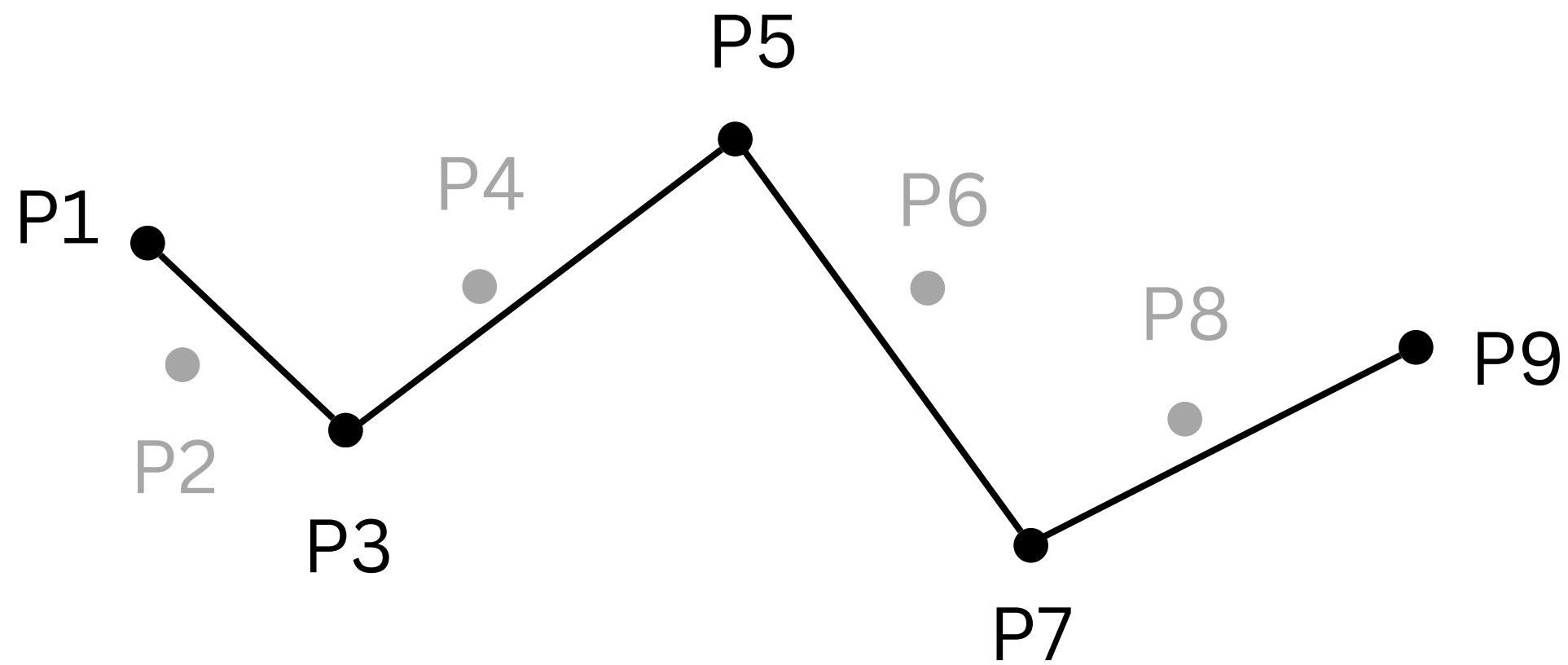


Simplify(P7, P9)
Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 **P7** P8 **P9**

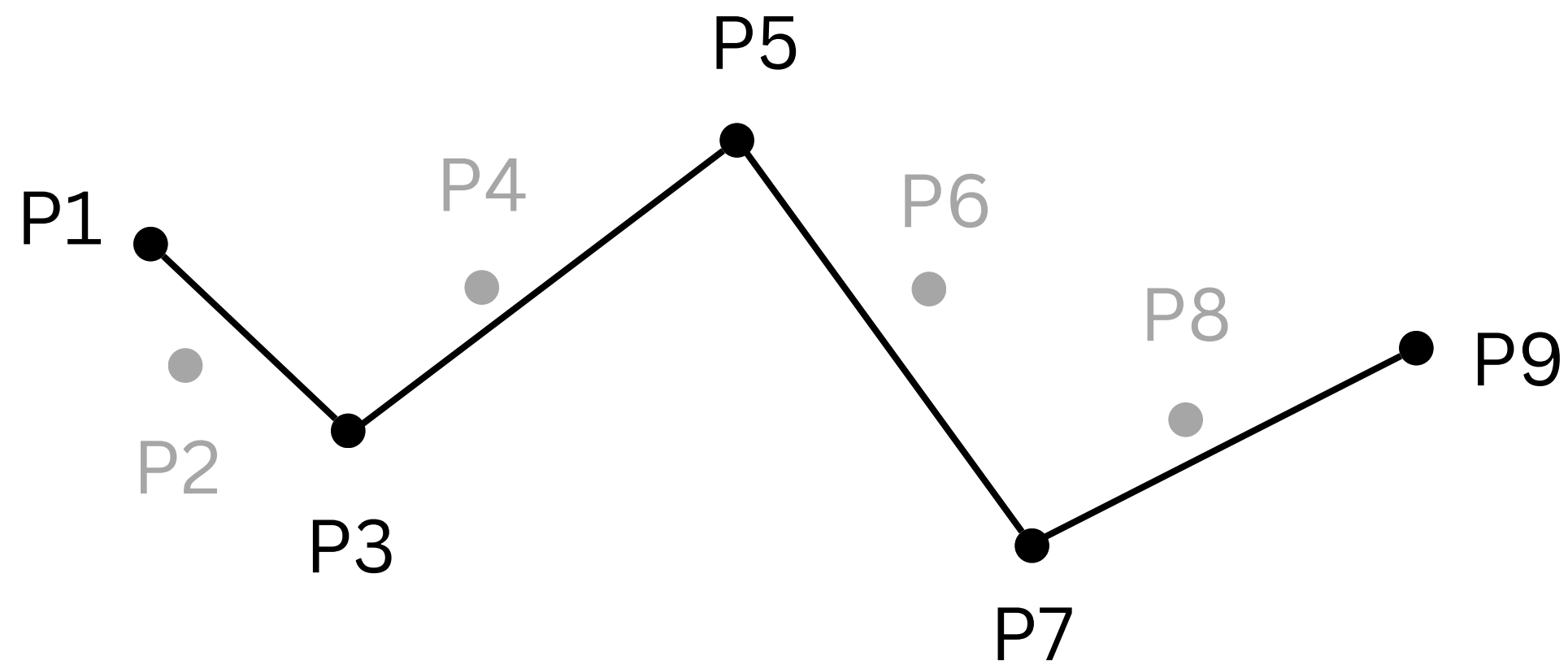


Simplify(P1, P9)

Call Stack

Array

P1 P2 P3 P4 P5 P6 P7 P8 **P9**



Call Stack

Array

P1 P2 P3 P4 P5 P6 P7 P8 P9

